Balochistan Board — Matric Physics

Unit 11: Sound — Numerical Problems

Statement:

The intensity of sound at a distance of 30 metres from a loudspeaker is 3.2 x 10^{-6} W m^{-2}. What will the intensity level in decibels? The threshold of hearing is 1 x 10^{-12} W m^{-2}.

Given:

Intensity (I) = 3.2 x 10^{-6} W m^{-2}

Threshold of hearing (I■) = 1 x 10^{-12} W m^{-2}

Formula:

$$\beta = 10 \log_{10} \left(\frac{I}{I_0} \right)$$

Solution:

$$\beta = 10 \log_{10} \left(\frac{3.2 \times 10^{-6}}{1 \times 10^{-12}} \right)$$

$$\beta = 10 \log_{10} (3.2 \times 10^{6})$$

$$\beta = 10 \times (6 + \log_{10} 3.2)$$

$$\beta = 10 \times (6 + 0.505)$$

$$\beta = 10 \times 6.505$$

$$\beta \approx 65 \text{ dB}$$

Answer: The intensity level is 65 dB.

Statement:

A vacuum cleaner is used to clean the carpet of a room. The intensity of sound produced by the vacuum cleaner is 1 x 10^{-5} W m^{-2}. Express the intensity level in decibels. The threshold of hearing i.e. faintest audible sound is 1 x 10^{-12} W m^{-2}.

Given:

Intensity (I) = 1 x 10^{-5} W m^{-2}

Threshold of hearing (I■) = 1 x 10^{-12} W m^{-2}

Formula:

$$\beta = 10\log_{10}\left(\frac{I}{I_0}\right)$$

Solution:

$$\beta = 10\log_{10}\left(\frac{1\times10^{-5}}{1\times10^{-12}}\right)$$

$$\beta = 10\log_{10}(10^{7})$$

$$\beta = 10\times7$$

$$\beta = 70\ \text{dB}$$

Answer: The intensity level is 70 dB.

Statement:

A person is standing in front of a high building. If he shouts loudly and hears the echo of his voice after one second, what is the distance of the building from him? Take speed of sound in air as 344 m s^{-1}.

Given:

Time for echo (t) = 1 s

Speed of sound (v) = 344 m s^{-1}

Formula:

$$d = \frac{v\times t}{2}$$

Note:

The total distance travelled by sound is 2d (to the building and back), so we divide by 2.

Solution:

$$d = \frac{344\times1}{2}$$

$$d = \frac{344}{2}$$

$$d = 172 \text{ m}$$

Answer: The distance of the building from the person is 172 m.

Statement:

A sound wave is travelling in air at room temperature with a wavelength of 17.2 cm. Calculate the frequency of the sound wave if speed of sound in air is 344 m s^{-1}.

Given:

Wavelength (λ) = 17.2 cm = 0.172 m

Speed of sound (v) = 344 m s^{-1}

Formula:

$$v = f\lambda \quad \Rightarrow \quad f = \frac{v}{\lambda}$$

Solution:

$$f = \frac{344}{0.172}$$

$$f = 2000 \text{ Hz}$$

Answer: The frequency of the sound wave is 2000 Hz.

Statement:

An ultrasound wave has a frequency of 35 kHz and travels in water. What will be its wavelength in water if speed of sound in water is 1400 m s^{-1}?

Given:

Frequency (f) = 35 kHz = 35,000 Hz

Speed of sound in water (v) = 1400 m s^{-1}

Formula:

$$\lambda = \frac{v}{f}$$

Solution:

$$\lambda = \frac{1400}{35000}$$

$$\lambda = 0.04 \text{ m}$$

Answer: The wavelength of the ultrasound wave in water is 0.04 m.

Statement:

A certain sound signal has a frequency of 8 kHz and wavelength 4.25 cm in air. Calculate the speed of sound in air.

Given:

Frequency (f) = 8 kHz = 8000 Hz

Wavelength (λ) = 4.25 cm = 0.0425 m

Formula:

$$v = f\lambda$$

Solution:

$$v = 8000 \times 0.0425$$

$$v = 340 \text{ m s}^{-1}$$

Answer: The speed of sound in air is 340 m s■¹.